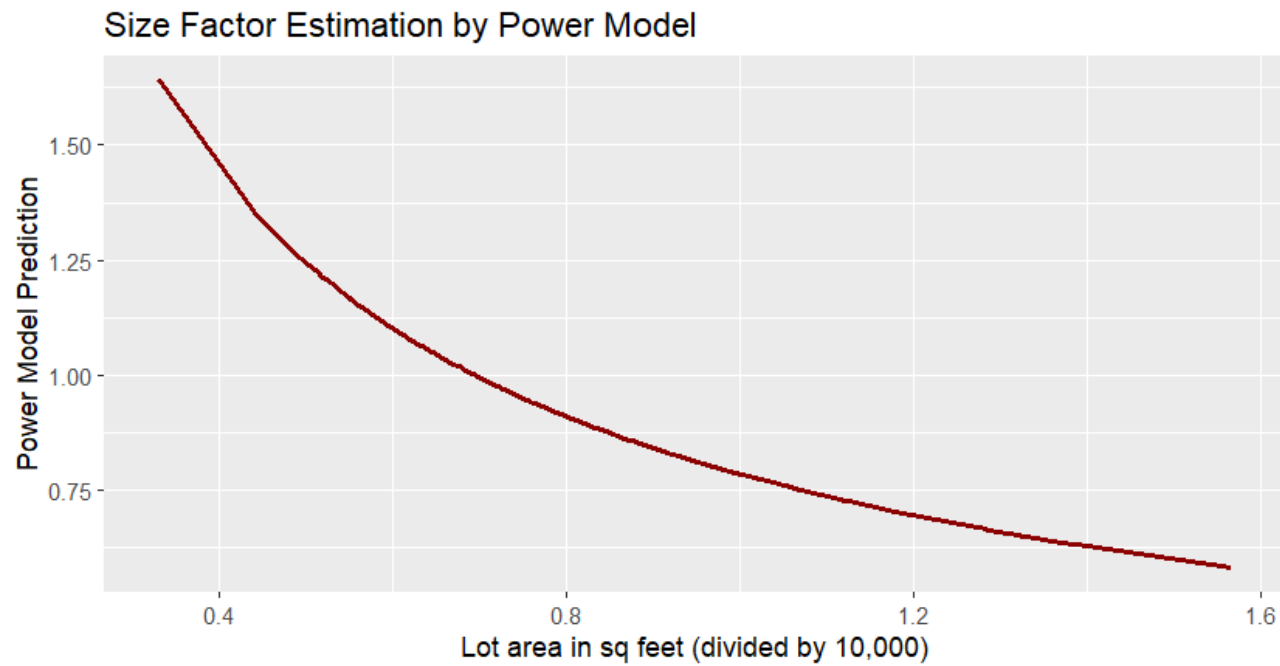


## Appendix

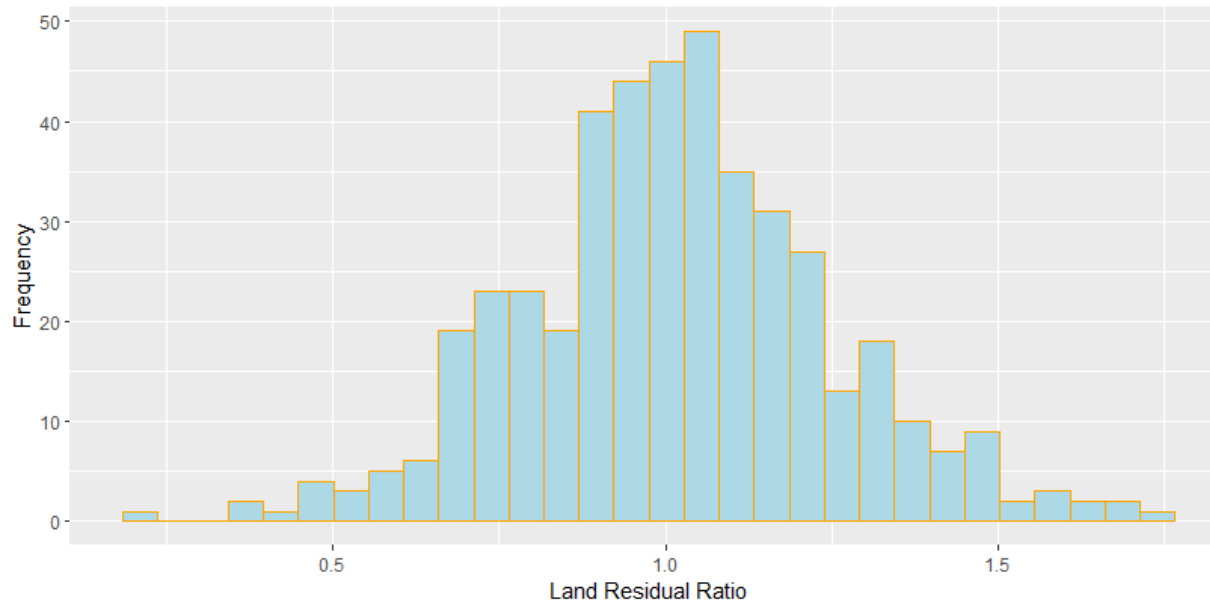
### Calculating Land Residual

*Fig A 1 Size Factor Estimation by Power Model*



## Unit Valuation Method

**Fig A 2. Land Residual Ratio after Adjustment for Land Features and Neighborhood**



**Table A 1 Mann-Whitney Results - general land value adjusted for culdesac**

| Mann-Whitney Results - general land value adjusted for culdesac |       |        |           |                   |
|-----------------------------------------------------------------|-------|--------|-----------|-------------------|
| Variable                                                        | Value | Number | Mean Rank | Ays. Significance |
| corner                                                          | 0     | 378    | 219       | 0.12              |
|                                                                 | 1     | 68     | 246       |                   |
| culdesac                                                        | 0     | 417    | 225       | 0.48              |
|                                                                 | 1     | 29     | 207       |                   |
| alley                                                           | 0     | 301    | 247       | 0.00              |
|                                                                 | 1     | 145    | 175       |                   |
| wooded                                                          | 0     | 410    | 225       | 0.35              |

|          |   |     |     |      |
|----------|---|-----|-----|------|
|          | 1 | 36  | 204 |      |
| pie_shpe | 0 | 420 | 221 | 0.12 |
|          | 1 | 26  | 261 |      |
| slope    | 0 | 389 | 225 | 0.53 |
|          | 1 | 57  | 213 |      |
| abv_road | 0 | 367 | 228 | 0.12 |
|          | 1 | 79  | 203 |      |

Table A 2 **Mann-Whitney Results - general land value adjusted for culdesac and alley**

| <b>Mann-Whitney Results - general land value adjusted for culdesac and alley</b> |              |               |                  |                          |
|----------------------------------------------------------------------------------|--------------|---------------|------------------|--------------------------|
| <b>Variable</b>                                                                  | <b>Value</b> | <b>Number</b> | <b>Mean Rank</b> | <b>Ays. Significance</b> |
| corner                                                                           | 0            | 378           | 220              | 0.19                     |
|                                                                                  | 1            | 68            | 242              |                          |
| culdesac                                                                         | 0            | 417           | 227              | 0.06                     |
|                                                                                  | 1            | 29            | 179              |                          |
| alley                                                                            | 0            | 301           | 221              | 0.60                     |
|                                                                                  | 1            | 145           | 228              |                          |
| wooded                                                                           | 0            | 410           | 226              | 0.11                     |
|                                                                                  | 1            | 36            | 191              |                          |
| pie_shpe                                                                         | 0            | 420           | 223              | 0.67                     |
|                                                                                  | 1            | 26            | 234              |                          |
| slope                                                                            | 0            | 389           | 226              | 0.37                     |
|                                                                                  | 1            | 57            | 209              |                          |
| abv_road                                                                         | 0            | 367           | 229              | 0.04                     |
|                                                                                  | 1            | 79            | 197              |                          |

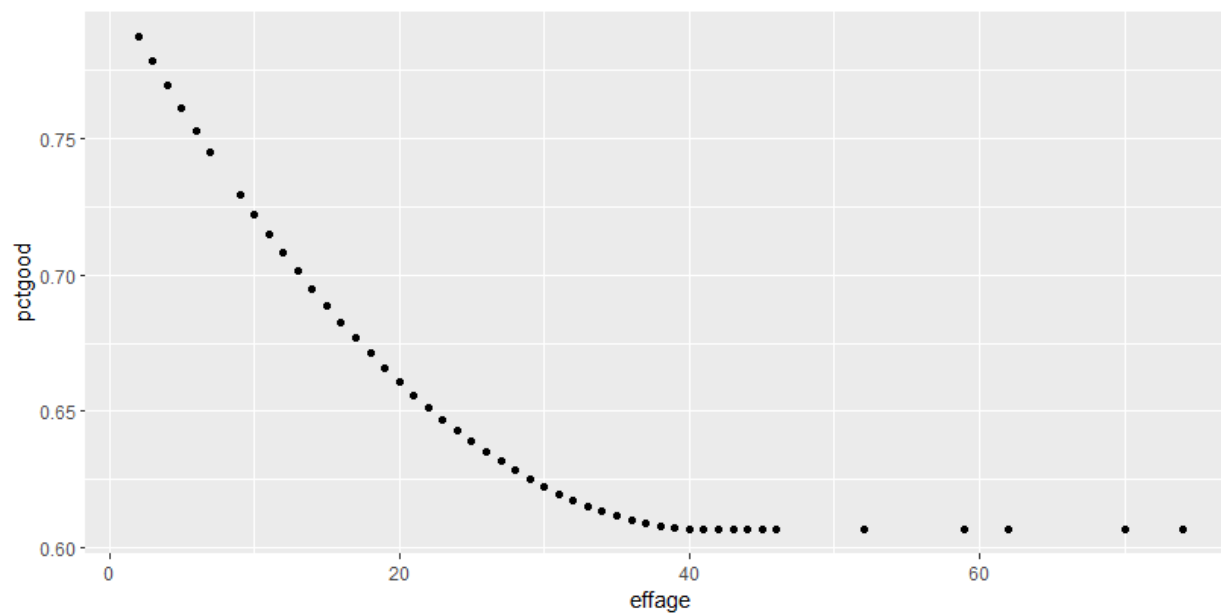
Table A 3 **Mann-Whitney Results - general land value adjusted for culdesac, alley, and abv\_road**

| <b>Mann-Whitney Results - general land value adjusted for culdesac, alley, and abv_road</b> |
|---------------------------------------------------------------------------------------------|
|---------------------------------------------------------------------------------------------|

| Variable | Value | Number | Mean Rank | Ays. Significance |
|----------|-------|--------|-----------|-------------------|
| corner   | 0     | 378    | 221       | 0.29              |
|          | 1     | 68     | 239       |                   |
| culdesac | 0     | 417    | 225       | 0.41              |
|          | 1     | 29     | 205       |                   |
| alley    | 0     | 301    | 223       | 0.84              |
|          | 1     | 145    | 225       |                   |
| wooded   | 0     | 410    | 227       | 0.09              |
|          | 1     | 36     | 189       |                   |
| pie_shpe | 0     | 420    | 222       | 0.47              |
|          | 1     | 26     | 241       |                   |
| slope    | 0     | 389    | 223       | 0.70              |
|          | 1     | 57     | 230       |                   |
| abv_road | 0     | 367    | 224       | 0.87              |
|          | 1     | 79     | 222       |                   |

## Market-based Depreciation Factor

**Fig A 3. Depreciation Factor versus Effective Age**



## Testing Building Characteristics

**Table A 4. Ratio Statistics for Mkfact1 adjusted for Split Level (Mkfactor2)**

| Split-level | Median | Lower Bound | Upper Bound | Actual Coverage | Minimum | Maximum | COD   |
|-------------|--------|-------------|-------------|-----------------|---------|---------|-------|
| 0           | 1.00   | 0.98        | 1.02        | 95.54           | 0.41    | 1.79    | 13.82 |
| 1           | 1.00   | 0.89        | 1.08        | 95.90           | 0.61    | 1.21    | 12.53 |

Table A 5. Ratio Statistics for Mkfact3 (adjusted by Split-level and Garage Type) based on Bedrooms

| Garage Type | Median | Lower Bound | Upper Bound | Actual Coverage | Minimum | Maximum | COD   |
|-------------|--------|-------------|-------------|-----------------|---------|---------|-------|
| 2           | 0.95   | 0.91        | 0.98        | 94.51           | 0.55    | 1.78    | 14.22 |
| 3           | 1.02   | 1           | 1.04        | 95.05           | 0.41    | 1.55    | 12.27 |
| 4           | 0.93   | 0.85        | 1.02        | 92.16           | 0.62    | 1.53    | 19.03 |
| 2           | 0.95   | 0.91        | 0.98        | 94.51           | 0.55    | 1.78    | 14.22 |

Table A 6. Ratio Statistics for Mkfact4 (adjusted by Split-level, Garage Type, and Bedrooms) based on Carport

| Garage Type | Median | Lower Bound | Upper Bound | Actual Coverage | Minimum | Maximum | COD   |
|-------------|--------|-------------|-------------|-----------------|---------|---------|-------|
| 2           | 0.95   | 0.91        | 0.98        | 94.51           | 0.55    | 1.78    | 14.22 |
| 3           | 1.02   | 1           | 1.04        | 95.05           | 0.41    | 1.55    | 12.27 |
| 4           | 0.93   | 0.85        | 1.02        | 92.16           | 0.62    | 1.53    | 19.03 |
| 2           | 0.95   | 0.91        | 0.98        | 94.51           | 0.55    | 1.78    | 14.22 |

Table A 7. Ratio Statistics for Mkfact5 (adjusted by Split-level, Garage Type, Bedrooms, and Carport) based on Family Room

| Garage Type | Median | Lower Bound | Upper Bound | Actual Coverage | Minimum | Maximum | COD   |
|-------------|--------|-------------|-------------|-----------------|---------|---------|-------|
| 0           | 0.95   | 0.93        | 0.98        | 94.33           | 0.39    | 1.85    | 15.16 |
| 1           | 1.03   | 1.01        | 1.05        | 95.09           | 0.6     | 1.63    | 11.59 |
| 2           | 1.01   | 0.97        | 1.05        | 93.61           | 0.71    | 1.19    | 8.15  |